



OBS & BROADCAST INTEGRATION SPECIFICATION

GameVerse — Esports Tournament Operations & Live Broadcast Platform

Version: 1.0 | June 2025 |  Broadcast Infrastructure Document

TABLE OF CONTENTS

1. Document Overview & Broadcast Philosophy
 2. Broadcast Architecture Overview
 3. OBS Integration Architecture
 4. Overlay System Overview
 5. Overlay Route Specification
 6. Overlay Type 01 — Leaderboard Overlay
 7. Overlay Type 02 — Match Information Overlay
 8. Overlay Type 03 — Team Statistics Overlay
 9. Overlay Type 04 — Player Statistics Overlay
 10. Overlay Type 05 — Sponsor Overlay
 11. Overlay Type 06 — Winner Overlay
 12. Overlay Live Data Update System
 13. Overlay Token & Security
 14. Overlay Customization & Theming
 15. Broadcast Producer Dashboard
 16. OBS WebSocket Integration
 17. Stream Health Monitoring
 18. Overlay Animation Specification
 19. Testing & QA for Overlays
 20. Troubleshooting Guide
-
-

SECTION 1 — DOCUMENT OVERVIEW & BROADCAST PHILOSOPHY

1.1 Purpose

This document defines the complete **OBS & Broadcast Integration Specification** for GameVerse — every overlay type, every live data update mechanism, every route, every animation, and every configuration option that powers professional esports broadcast production on the platform.

1.2 Broadcast Philosophy

BROADCAST DESIGN PRINCIPLES:

1. ZERO MANUAL DATA ENTRY

Broadcast producers never type scores, team names, or stats.
All overlay data flows automatically from the scoring engine.

2. ALWAYS SHOW SOMETHING

Overlays never go blank. On connection loss, they freeze at the last valid state. Blank = broken on live broadcast.

3. PRODUCTION-GRADE ANIMATIONS

Every data change animates smoothly. Rank changes, score updates, and transitions feel broadcast-quality, not amateur.

4. BROWSER SOURCE NATIVE

All overlays are designed specifically for OBS Browser Source. Transparent backgrounds. No scroll. No external fonts loaded. Offline-tolerant with SockJS fallback.

5. ORGANIZER-BRANDABLE

Every overlay inherits the organization's brand colors, fonts, and logo. White-label ready with one configuration.

1.3 Supported OBS Versions

COMPATIBILITY:

- OBS Studio: 28.x, 29.x, 30.x (all current versions)
- Streamlabs Desktop: 1.x, 2.x (Browser Source compatible)
- vMix: Any version with Browser Input support
- XSplit: Broadcaster with webpage source support

Browser Engine: Chromium (embedded in OBS)

- Supports: WebSocket, SockJS, CSS animations, WebP images
- Supports: CSS variables, CSS Grid, Flexbox
- Does NOT support: Certain Web APIs (getUserMedia, etc.)
→ Not needed for overlays

1.4 Overlay Dimensions Reference

STANDARD OVERLAY DIMENSIONS:

- Full HD (1080p): 1920 × 1080 pixels ← Primary target
QHD (1440p): 2560 × 1440 pixels ← Supported
4K (2160p): 3840 × 2160 pixels ← Supported (scale factor)

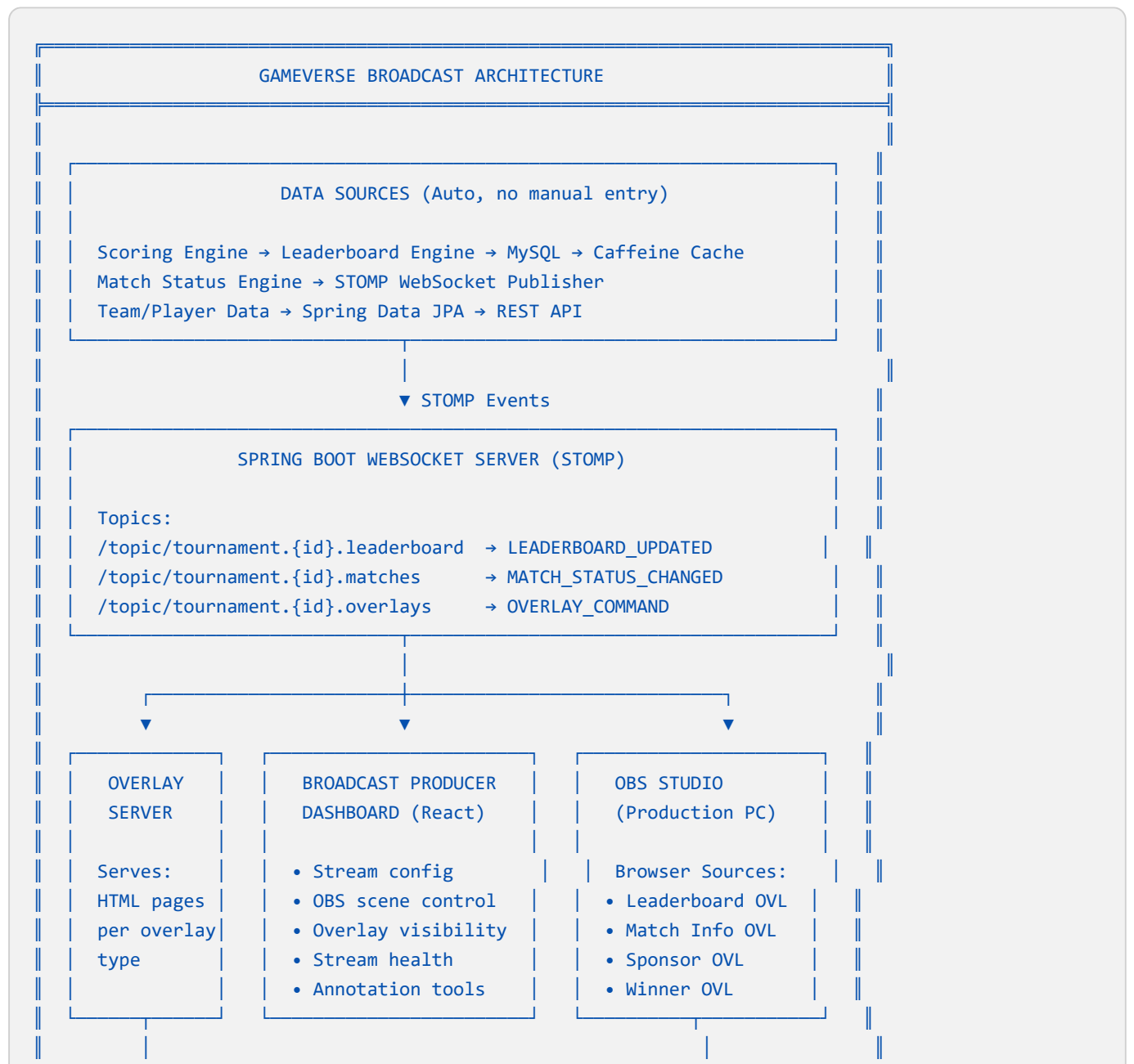
OVERLAY-SPECIFIC DIMENSIONS:

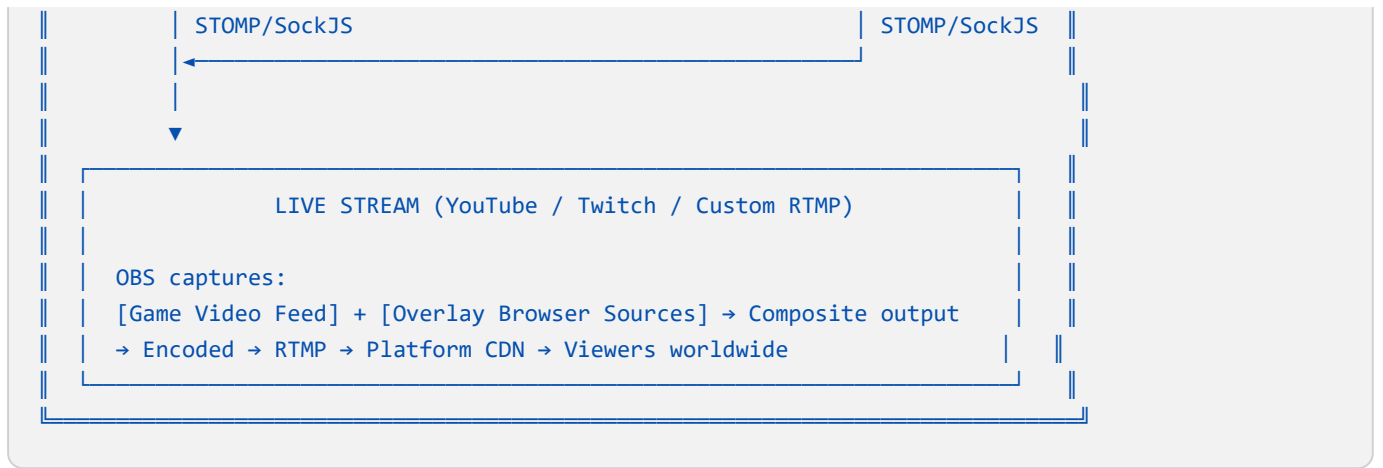
Overlay Type	Default Size	OBS Layer
--------------	--------------	-----------

Leaderboard (Full)	420 × 900 px	Right panel
Leaderboard (Compact/10)	380 × 600 px	Right panel
Match Information Bar	1920 × 80 px	Bottom strip
Team Statistics	480 × 320 px	Corner panel
Player Statistics	360 × 480 px	Side panel
Sponsor Banner	1920 × 120 px	Top/bottom bar
Sponsor Logo (corner)	200 × 100 px	Corner
Winner Celebration	1920 × 1080 px	Full screen
Winner Podium	800 × 400 px	Center stage

SECTION 2 — BROADCAST ARCHITECTURE OVERVIEW

2.1 Complete Broadcast System Map





2.2 Data Flow: Score Update → Overlay Visible

SCORE → OVERLAY UPDATE FLOW:

```
[Referee submits result]
  ↓ REST API POST
[Spring Boot validates + saves to MySQL]
  ↓ @TransactionalEventListener
[LeaderboardEngine.recalculate()]
  ↓ Updated leaderboard saved
[SimpMessagingTemplate.convertAndSend(
  "/topic/tournament.{id}.leaderboard",
  LEADERBOARD_UPDATED payload
)]
  ↓ STOMP broker routes
[OBS Browser Source receives STOMP frame]
  ↓ JavaScript event handler
[updateLeaderboardDOM(newData)]
  ↓ DOM manipulation
[CSS transition animates rows to new positions]
  ↓
OVERLAY VISUALLY UPDATED ON STREAM
```

Total time: ~200-500ms from result submission
Target: < 3 seconds ☒

SECTION 3 — OBS INTEGRATION ARCHITECTURE

3.1 OBS Browser Source Configuration

HOW BROADCAST PRODUCERS ADD OVERLAYS TO OBS:

Step 1: Open Broadcast Dashboard in GameVerse
(Command Center → Broadcast → Overlays tab)

Step 2: Find desired overlay type

Each overlay shows:

- └─ Preview thumbnail
- └─ Copy URL button
- └─ Recommended OBS dimensions
- └─ Configuration options

Step 3: In OBS Studio:

Sources panel → (+) Add → Browser

Step 4: Browser Source Settings:

URL:	https://gameverse.gg/overlay/ tournament/{id}/leaderboard? token={overlayToken}
Width:	420
Height:	900
☒	Shutdown source when not visible
☒	Refresh browser when scene becomes active
Custom CSS: (leave empty – overlay handles it)	

Step 5: Position overlay in OBS scene layout

- └─ Drag to desired position
- └─ Use OBS alignment guides
- └─ Lock layer when positioned

Step 6: Verify in Broadcast Dashboard:

- └─ Overlay shows "● Connected" status
- └─ Preview pane shows live data
- └─ Test animation by toggling visibility

3.2 Recommended OBS Scene Layout

RECOMMENDED OBS SCENE STRUCTURE FOR GAMEVERSE TOURNAMENT:

Scene: "Match Live" (primary scene)

OBS SOURCE STACK (bottom to top):
Layer 1 (bottom): Game Video Capture / NDI Source
Layer 2: OVL-05 Sponsor Banner (top or bottom bar)
Layer 3: OVL-02 Match Information Bar (bottom strip)
Layer 4: OVL-01 Leaderboard Overlay (right side)
Layer 5 (top): OVL-WINNER (hidden by default, shown on finale)

Scene: "Break Screen"

Layer 1: Tournament graphic (static image)
Layer 2: OVL-05 Sponsor Banner (rotating sponsors)

Layer 3: OVL-02 Match Info Bar (shows next match countdown)

Scene: "Result Reveal"

Layer 1: Game Video Capture

Layer 2: OVL-01 Full Leaderboard (center, full-width variant)

Layer 3: OVL-03 Team Statistics (featured team)

Scene: "Grand Finale"

Layer 1: Tournament closing graphic

Layer 2: OVL-06 Winner Overlay (full screen)

Scene: "Player Spotlight"

Layer 1: Game Video Capture

Layer 2: OVL-04 Player Statistics (featured player)

Layer 3: OVL-05 Sponsor Logo (corner)

SECTION 4 — OVERLAY SYSTEM OVERVIEW

4.1 All Overlay Types

GAMEVERSE OVERLAY CATALOG:

ID	Overlay Name	Trigger	Auto-Update Source
01	Leaderboard (Full, 16-team)	Always visible	LEADERBOARD_UPDATED
01B	Leaderboard (Compact, Top10)	Always visible	LEADERBOARD_UPDATED
01C	Leaderboard (Mini, Top5)	Always visible	LEADERBOARD_UPDATED
02	Match Information Bar	Always visible	MATCH_STATUS_CHANGED
02B	Match Countdown Timer	Scheduled	MATCH_STATUS_CHANGED
03	Team Statistics Panel	Producer ctrl	RESULT_PUBLISHED
03B	Team vs Team Comparison	Producer ctrl	RESULT_PUBLISHED
04	Player Statistics Card	Producer ctrl	RESULT_PUBLISHED
04B	Player Kill Feed	Producer ctrl	RESULT_PUBLISHED
05	Sponsor Banner (Full-width)	Always visible	Static (rotates)
05B	Sponsor Logo (Corner)	Always visible	Static
06	Winner Celebration (Full)	Tournament end	TOURNAMENT_COMPLETED
06B	Winner Podium (Compact)	Tournament end	TOURNAMENT_COMPLETED
06C	Result Splash (Per-match)	Per result	RESULT_PUBLISHED

4.2 Overlay Technology Stack

OVERLAY TECHNICAL STACK:

RENDERING: Pure HTML5 + CSS3 + Vanilla JavaScript

- └ NO React framework (too heavy for OBS browser source)
- └ NO external JavaScript libraries in runtime
- └ NO external CSS frameworks (no Tailwind CDN)
- └ All fonts: Embedded as base64 in CSS (no font CDN calls)
- └ All icons: Inline SVG (no icon library CDN)

REAL-TIME: @stomp/stompjs (bundled, minified)

- └ Connected to Spring Boot WebSocket STOMP endpoint
- └ SockJS fallback: bundled sockjs.min.js
- └ Total bundle size: < 40KB (lightweight for OBS)

ANIMATION: CSS Transitions + CSS Animations (GPU-accelerated)

- └ transform: translateY (row repositioning)
- └ opacity transitions (fade in/out)
- └ @keyframes for celebration effects
- └ No JavaScript animation libraries (no GSAP, no Anime.js)

FONTS (embedded):

- └ Primary: Rajdhani (esports aesthetic, semi-condensed)
- └ Secondary: Inter (data readability)
- └ Monospace: JetBrains Mono (UIDs, codes)

DELIVERY:

- └ Static HTML served by Nginx (5-minute edge cache)
- └ Spring Boot injects initial state into HTML template
- └ JavaScript connects STOMP for live updates
- └ No server-side rendering on update – all client-side DOM

SECTION 5 — OVERLAY ROUTE SPECIFICATION

5.1 Complete Route Map

OVERLAY URL STRUCTURE:

Base: `https://gameverse.gg/overlay/tournament/{tournamentId}/{overlayType}`

Query Parameters:

└ token={overlayToken}	REQUIRED: Security token (from Broadcast Dashboard)
└ theme={themeName}	OPTIONAL: dark (default) light custom
└ scale={number}	OPTIONAL: 1.0 (default), 1.25, 1.5, 2.0
└ debug={boolean}	OPTIONAL: true shows connection status overlay

COMPLETE ROUTE TABLE:

--

Route	Overlay Type
/overlay/tournament/{id}/leaderboard	OVL-01: Full Leaderboard
/overlay/tournament/{id}/leaderboard/top10	OVL-01B: Top 10
/overlay/tournament/{id}/leaderboard/top5	OVL-01C: Top 5 Mini
/overlay/tournament/{id}/match-info	OVL-02: Match Info Bar
/overlay/tournament/{id}/match-countdown	OVL-02B: Match Countdown
/overlay/tournament/{id}/team/{teamId}/stats	OVL-03: Team Statistics
/overlay/tournament/{id}/team/compare	OVL-03B: Team vs Team
/overlay/tournament/{id}/player/{userId}/stats	OVL-04: Player Stats
/overlay/tournament/{id}/player/killfeed	OVL-04B: Kill Feed
/overlay/tournament/{id}/sponsor	OVL-05: Sponsor Banner
/overlay/tournament/{id}/sponsor/corner	OVL-05B: Sponsor Corner
/overlay/tournament/{id}/winner	OVL-06: Winner Full
/overlay/tournament/{id}/winner/podium	OVL-06B: Winner Podium
/overlay/tournament/{id}/result-splash	OVL-06C: Result Splash

5.2 Route Handler Specification (Spring Boot)

OVERLAY ROUTE HANDLER:

GET /overlay/tournament/{tournamentId}/{overlayType}
(+ sub-paths for variants)

Spring Controller: OverlayController.java

Processing Steps:

1. Extract overlay token from query param
2. Validate token:
 - └ Token exists in overlay_configs table
 - └ tournament_id matches route param
 - └ overlay_type matches route param
 - └ token not expired (tournament_end + 30 days)
 - └ If invalid: Return 401 Unauthorized HTML error page
3. Fetch initial tournament state:
 - └ Tournament basic info (name, game, status)
 - └ Current leaderboard (from Caffeine cache)
 - └ Current active match status
 - └ Org branding (colors, logo, fonts)
 - └ Sponsor data (for sponsor overlays)
4. Render HTML template:
 - └ Inject initial state as window.__INITIAL_STATE__ JSON
 - └ Inject STOMP endpoint URL and overlay token
 - └ Inject org branding as CSS variables
 - └ Return complete HTML page
5. Response headers:
 - └ Cache-Control: public, max-age=300 (5 minutes)
 - └ Content-Type: text/html; charset=UTF-8
 - └ X-Frame-Options: SAMEORIGIN (OBS is same-origin compatible)

5.3 Overlay HTML Template Structure

```

<!-- OVERLAY PAGE STRUCTURE (served by Spring Boot) -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>GameVerse Overlay – {overlayType}</title>

  <!-- ALL styles inline – no external CSS requests -->
  <style>
    /* RESET */
    * { margin: 0; padding: 0; box-sizing: border-box; }

    /* TRANSPARENT BACKGROUND – critical for OBS compositing */
    html, body {
      background: transparent !important;
      overflow: hidden;
      width: 100%;
      height: 100%;
    }

    /* BRAND CSS VARIABLES – injected by Spring Boot from org config */
    :root {
      --brand-primary:    {org.primaryColor};
      --brand-secondary:  {org.secondaryColor};
      --brand-accent:     {org.accentColor};
      --brand-bg:         {org.overlayBgColor};
      --brand-text:       {org.textColor};
      --brand-font:       '{org.fontFamily}', 'Rajdhani', sans-serif;
    }

    /* OVERLAY-SPECIFIC STYLES (per overlay type, also inline) */
    /* ... all CSS embedded, no @import ... */
  </style>
</head>
<body>

  <!-- OVERLAY ROOT – JavaScript renders into this -->
  <div id="overlay-root">
    <!-- Server-side initial render (prevents flash of empty content) -->
    <!-- Injected by Spring Boot Thymeleaf template -->
  </div>

  <!-- CONNECTION STATUS DOT (configurable to hide) -->
  <div id="connection-dot" class="status-dot disconnected"></div>

  <!-- INITIAL STATE – injected by Spring Boot, eliminates first-load flash -->
  <script>
    window.__INITIAL_STATE__ = /*[[$${initialStateJson}]]*/ {};
    window.__CONFIG__ = {
      stompUrl:      'wss://api.gameverse.gg/ws',
      sockjsUrl:     'https://api.gameverse.gg/ws',
      overlayToken:  /*[[$${overlayToken}]]*/ '',
      tournamentId: /*[[$${tournamentId}]]*/ '',
      overlayType:   /*[[$${overlayType}]]*/ '',
    };
  </script>

```

```
    debug:      /*[[${debug}]]*/ false
  };
</script>

<!-- STOMP RUNTIME – bundled, minified, no CDN -->
<script src="/overlay/static/stomp.min.js"></script>
<script src="/overlay/static/sockjs.min.js"></script>

<!-- OVERLAY ENGINE – shared connection + event handling -->
<script src="/overlay/static/overlay-engine.js"></script>

<!-- OVERLAY RENDERER – specific to overlay type -->
<script src="/overlay/static/{overlayType}-renderer.js"></script>

</body>
</html>
```

SECTION 6 — OVERLAY TYPE 01: LEADERBOARD OVERLAY

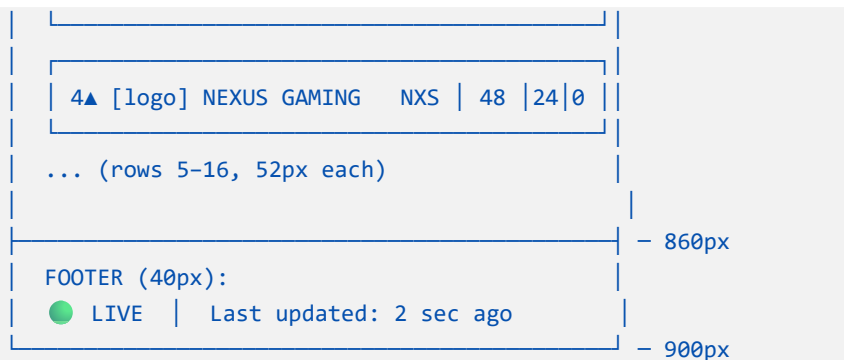
6.1 Overview

OVERLAY: OVL-01 – Full Leaderboard
Route: /overlay/tournament/{id}/leaderboard
Size: 420 × 900 px (default) | Variants: Top10 (380×600), Top5 (320×400)
Purpose: Shows live tournament standings during match broadcast
Updates: On every LEADERBOARD_UPDATED STOMP event

6.2 Visual Layout Specification

OVL-01: FULL LEADERBOARD (420 × 900 px)

					– 0px
[ORG LOGO] BGMI WEEKEND CUP #12 Round 2 of 4 • Match 8/16					← Header (60px)
					– 60px
COLUMN HEADERS (28px): # TEAM PTS KILLS CD					
					– 88px
1▲ [logo]	STORM SQUAD	STM	62	38 2	← Row (52px each)
2▼ [logo]	HYDRA ESPORTS	HYD	58	31 1	
3– [logo]	PHOENIX RISING	PHX	51	27 1	



ROW ANATOMY (52px height, 420px width):

[#]	[▲▼]	[16×16 logo]	[TEAM NAME]	[TAG]	[PTS]	[KLS]	[CD]
6px	20px	flex-grow	36px	48px	36px	24px	

RANK CHANGE INDICATORS:

- ▲ (up arrow, green) = moved up since last match
- ▼ (down arrow, red) = moved down since last match
- (dash, grey) = no change
- ★ (star, gold) = CHICKEN DINNER in last match

6.3 Leaderboard Data Schema (from STOMP)

DATA CONSUMED FROM: LEADERBOARD_UPDATED event
(Section 5.2 of Real-Time Architecture Document)

FIELDS USED BY LEADERBOARD OVERLAY:

Per team row:

- rank → Rank number (1, 2, 3...)
- rankChangeDirection → "UP" | "DOWN" | "SAME"
- teamName → "Storm Squad"
- teamTag → "STM"
- logoUrl → CDN URL for team logo (16×16 in overlay)
- totalPoints → 62.0 → display as "62"
- totalKills → 38
- chickenDinners → 2 (CD column)
- isDisqualified → If true: show ⚠️ DQ badge

Tournament header:

- tournamentName → "BGMI Weekend Cup #12"
- matchesCompleted → 8
- totalMatches → 16 → display "Match 8/16"
- currentRound → 2
- totalRounds → 4 → display "Round 2 of 4"

6.4 Leaderboard Animation Specification

ANIMATION BEHAVIOR ON LEADERBOARD_UPDATED:

STEP 1: Receive STOMP event (LEADERBOARD_UPDATED)
STEP 2: Compare new ranking order with current DOM order
STEP 3: Flash animation on changed cells:
→ Points that changed: background flash yellow → normal (300ms)
→ New kills: same yellow flash
STEP 4: Calculate new vertical positions for all rows
STEP 5: Apply CSS transitions for row repositioning:
→ All rows: transition: transform 600ms cubic-bezier(0.4, 0, 0.2, 1)
→ Rows animate to new Y positions simultaneously
→ Ascending team: translateY moves UP, color pulse green
→ Descending team: translateY moves DOWN, color pulse red
STEP 6: After animation completes (600ms):
→ Update rank numbers
→ Update rank change arrows (▲ ▼ -)
→ Update "Last updated: X sec ago" footer

CHICKEN DINNER ANIMATION:

When isChickenDinner changes to true for a match:

- Row: Gold border glow animation (1.5s)
- ★ star icon: spin animation (500ms)
- Row background: brief gold highlight (800ms)
- Then: returns to normal styled row

DISQUALIFICATION:

When isDisqualified becomes true:

- Row: red strikethrough effect on team name
- DQ badge appears: ⚠ DQ (red pill badge)
- Row moves to bottom of leaderboard
- Points displayed in red: "38pts (DQ)"

INITIAL LOAD (no animation):

- window.__INITIAL_STATE__ pre-populates all 16 rows
- Renders instantly with correct data
- No "loading" state visible
- First STOMP update: animates from there

6.5 Leaderboard Variants

OVL-01B: TOP 10 LEADERBOARD

Route: /overlay/tournament/{id}/leaderboard/top10
Size: 380 × 600 px
Changes: Only shows rank 1-10, rows slightly smaller (42px)
Use: When full 16-team list is too tall for scene layout

OVL-01C: TOP 5 MINI LEADERBOARD

Route: /overlay/tournament/{id}/leaderboard/top5
Size: 320 × 340 px
Changes: Only shows rank 1-5, compact rows (40px)
No kill/CD columns (points only)
Use: Corner placement, secondary stream (mobile stream)

SECTION 7 — OVERLAY TYPE 02: MATCH INFORMATION OVERLAY

7.1 Overview

OVERLAY: OVL-02 – Match Information Bar

Route: /overlay/tournament/{id}/match-info

Size: 1920 × 80 px (full-width bottom bar)

Purpose: Shows current match status, round info, and timing

Updates: On MATCH_STATUS_CHANGED STOMP event

7.2 Visual Layout Specification

OVL-02: MATCH INFORMATION BAR (1920 × 80 px)

[TOURNAMENT NAME]	[ROUND]	[MATCH STATUS]	[MATCH TIMER]
BGMI Weekend Cup #12	Round 2/4	● IN PROGRESS	🕒 12:34
	Match 3/16	[green pulse dot]	

SECTIONS (left to right):

- LEFT SECTION (480px):
 - Tournament name: "BGMI Weekend Cup #12"
 - Organization: "by Hydra Events" (small, below)
- CENTER-LEFT (320px):
 - Round: "Round 2 of 4"
 - Match: "Match 3 of 16"
- CENTER (480px):
 - Status: ● IN PROGRESS (pulsing green dot)
 - OR: II PAUSED (pulsing orange)
 - OR: LOBBY OPEN (static blue)
 - OR: COMPLETED (static grey)
 - Referee name: "Ref: Mike J." (small, below)
- RIGHT SECTION (640px):
 - Match timer: 🕒 12:34 (counting up from match start)
 - OR: Next match in: 08:42 (countdown to next match)
 - Scheduled time: "Scheduled: 2:30 PM IST" (small, below)

STATUS COLORS:

- SCHEDULED: Blue background pill #3B82F6
- LOBBY_OPEN: Cyan background pill #06B6D4
- IN_PROGRESS: Green background pill #10B981 (pulsing animation)
- PAUSED: Orange background pill #F59E0B (pulsing animation)

└─ RESULT_SUBMITTED: Purple background #8B5CF6
└─ COMPLETED: Grey background pill #6B7280

7.3 Match Information Data Schema

DATA CONSUMED FROM: MATCH_STATUS_CHANGED event

FIELDS USED:

└─ tournamentName → "BGMI Weekend Cup #12"
└─ orgName → "Hydra Events"
└─ matchNumber → 3
└─ totalMatches → 16
└─ roundNumber → 2
└─ totalRounds → 4
└─ currentStatus → "IN_PROGRESS"
└─ actualStartTime → ISO timestamp (for elapsed timer calculation)
└─ scheduledStart → ISO timestamp
└─ assignedReferee.username → "RefMike"
└─ estimatedEndTime → ISO timestamp (optional, if available)

TIMER LOGIC:

In Progress: elapsed = now - actualStartTime → display MM:SS (counting up)
Paused: elapsed frozen at pause time, "II 12:34" shown
Scheduled: countdown = scheduledStart - now → "Starts in MM:SS"
Completed: match duration shown → "Duration: 34:21"

7.4 Match Information Variant

OVL-02B: MATCH COUNTDOWN TIMER

Route: /overlay/tournament/{id}/match-countdown

Size: 400 × 200 px

Purpose: Standalone countdown to next match (used in break screens)

Updates: On MATCH_STATUS_CHANGED

VISUAL:

```
|      NEXT MATCH STARTS IN      |
|                                |
|      08 : 42 : 19              |
|      HH   MM   SS              |
|                                |
| Round 2 of 4 • Match 4 of 16  |
```

Behavior:

└─ Timer counts down to next scheduled match
└─ At 0:00: transitions to "MATCH STARTING" state
└─ When match starts: transitions to "MATCH IN PROGRESS" display
└─ Pulses red when under 60 seconds remaining

SECTION 8 — OVERLAY TYPE 03: TEAM STATISTICS OVERLAY

8.1 Overview

OVERLAY: OVL-03 – Team Statistics Panel

Route: /overlay/tournament/{id}/team/{teamId}/stats

Size: 480 × 320 px

Purpose: Feature a specific team's performance stats

Updates: On RESULT_PUBLISHED STOMP event

Control: Broadcast Producer selects featured team in Broadcast Dashboard

8.2 Visual Layout Specification

OVL-03: TEAM STATISTICS PANEL (480 × 320 px)

[TEAM LOGO 48×48] HYDRA ESPORTS HYD

Rank #2

TOURNAMENT STATS (this tournament)

POINTS	KILLS	CDs	MATCHES
58.0	31	1	8

LAST MATCH PERFORMANCE

Match 8 • Placement: 3rd • Kills: 5

Points: 15.0

MATCH HISTORY (sparkline):

[1st][3rd][2nd][5th][4th][2nd][1st][3rd]

← Team Header (72px)

← Stat row 1

← Last match

← Placement chart

– 0px

– 72px

– 320px

STAT CARD VALUES:

- POINTS: Sum of all total_points across published matches
- KILLS: Sum of all raw_kills across published matches
- CDs: Count of chicken dinners (1st place finishes)
- MATCHES: Count of completed matches

8.3 Dynamic Team Selection

DYNAMIC TEAM URL:

/overlay/tournament/{id}/team/{teamId}/stats

teamId can be:

- Specific UUID: /team/team-uuid-abc/stats
 - Always shows this specific team
 - Producer sets up per team in OBS
- Dynamic: /team/featured/stats
 - Shows whichever team the producer has "Featured"
 - Broadcast Dashboard: Team Spotlight selector
 - Producer picks team → OVERLAY_COMMAND sent
 - Overlay switches to featured team

FEATURED TEAM CONTROL:

Producer Dashboard → Team Spotlight → Select "Hydra Esports"

```
→ OVERLAY_COMMAND {  
  command: "UPDATE_FEATURED_TEAM",  
  overlayType: "TEAM_STATS",  
  params: { teamId: "team-uuid-hyd", teamName: "Hydra Esports" }  
}  
→ Overlay JS receives command → fetches team data → updates display
```

8.4 Team vs Team Comparison Variant

OVL-03B: TEAM VS TEAM COMPARISON

Route: /overlay/tournament/{id}/team/compare
?teamA={teamId}&teamB={teamId}

Size: 800 × 300 px

Purpose: Side-by-side stat comparison of two featured teams

Updates: On RESULT_PUBLISHED

VISUAL:

[HYD LOGO] HYDRA ESPORTS		VS	STORM SQUAD [STM LOGO]
Rank:	#2		#1
Points:	58.0		62.0
Kills:	31		38
CDs:	1		2
Avg Pos:	3.9		3.2
Best:	1st		1st

Bar widths are proportional (e.g., kills: 31 vs 38 → 44% vs 56% of bar)

Leading team's bar is brand accent color, trailing is muted.

SECTION 9 — OVERLAY TYPE 04: PLAYER STATISTICS OVERLAY

9.1 Overview

OVERLAY: OVL-04 – Player Statistics Card

Route: /overlay/tournament/{id}/player/{userId}/stats

Size: 360 × 480 px

Purpose: Spotlight individual player performance stats

Updates: On RESULT_PUBLISHED STOMP event

Control: Broadcast Producer selects featured player in Broadcast Dashboard

9.2 Visual Layout Specification

OVL-04: PLAYER STATISTICS CARD (360 × 480 px)

[AVATAR 64×64]

ARJUN SHARMA

ArjunOP

[TEAM LOGO 20×20]

Hydra Esports

IN-GAME NAME: ArjunOP

BGMI ID: 549****567 (masked)

TOURNAMENT STATS:

TOTAL	AVG KILLS	BEST GAME
KILLS: 18	2.25/mtch	6 kills

MATCHES	AVG PLACE	CDs
8	3.9	1

MATCH-BY-MATCH KILLS:

M1

M2

M3

M4

M5

M6

M7

M8

6

2

5

1

3

2

6

2

– 0px

← Player Header (88px)

– 88px

← UID display (36px)

– 124px

← Stat row (80px)

← Stat row (80px)

– 364px

← Kill chart (80px)

– 480px

DATA SOURCE:

- Player info: users table + team_members
- In-game UID: registration_rosters.in_game_uid (masked in display)
- Tournament stats: Aggregated from team_match_results
- (Note: Individual player kill tracking not supported in BGMI)

| without game API – shows team kill stats attributed to player)
└─ Match history: team_match_results per match

9.3 Player Kill Feed Variant

OVL-04B: PLAYER KILL FEED (Conceptual – Manual Entry)

Route: /overlay/tournament/{id}/player/killfeed

Size: 400 × 200 px

NOTE: Since GameVerse has no direct BGMI game API access, kill feed data is manually curated by Broadcast Producer or pulled from post-match result data.

VISUAL:

KILL FEED		Match 3 • LIVE
	ArjunOP (HYD) eliminated PhxSniper	
	StormIGL (STM) eliminated NxsPlayer3	
	Storm Squad – CHICKEN DINNER –	

Implementation: Broadcast Producer manually inputs kill events in Broadcast Dashboard annotation tool. Each annotation pushed via OVERLAY_COMMAND to this overlay.

SECTION 10 — OVERLAY TYPE 05: SPONSOR OVERLAY

10.1 Overview

OVERLAY: OVL-05 – Sponsor Banner

Route: /overlay/tournament/{id}/sponsor

Size: 1920 × 120 px (full-width banner)

Purpose: Display tournament sponsor branding during broadcast

Updates: Static (rotates between sponsors on timer)

Refreshes on OVERLAY_DATA_REFRESH command

10.2 Visual Layout Specification

OVL-05: SPONSOR BANNER (1920 × 120 px)

SINGLE SPONSOR MODE:

PRESENTED BY	[SPONSOR LOGO 300×80]
"Powering Esports Champions"	https://techbrandpro.com

MULTI-SPONSOR ROTATION MODE (3 sponsors):

Sponsor A displays for 15 seconds

→ Fade out transition (500ms)

→ Sponsor B fades in (500ms)

→ Sponsor B displays for 15 seconds

→ Fade to Sponsor C... → Loops

SPONSOR TIER DISPLAY:

└─ TITLE SPONSOR ("Presented by"):

└─ └─ Logo: 300×80px, "PRESENTED BY" label, tagline shown

└─ GOLD SPONSOR ("Supported by"):

└─ └─ Logo: 200×60px, smaller display

└─ COMMUNITY SPONSOR ("Community Partner"):

└─ └─ Logo: 160×50px, text only or small logo

SPONSOR CORNER VARIANT (OVL-05B):

Route: /overlay/tournament/{id}/sponsor/corner

Size: 200 × 100 px

Content: Just sponsor logo + "Partner" label

Use: Always-on corner placement during match live scenes

10.3 Sponsor Data Schema

DATA SOURCE:

Spring Boot fetches from tournament_sponsors + sponsors tables

```
window.__INITIAL_STATE__.sponsors = [
  {
    "sponsorId": "sponsor-uuid-1",
    "companyName": "TechBrand Pro",
    "logoUrl": "https://cdn.gameverse.gg/sponsors/techbrand.webp",
    "tagline": "Powering Esports Champions",
    "websiteUrl": "https://techbrandpro.com",
    "tier": "TITLE",
    "displayDurationSeconds": 15,
    "displayOrder": 1
  },
  {
    "sponsorId": "sponsor-uuid-2",
    "companyName": "GearMax",
    "logoUrl": "https://cdn.gameverse.gg/sponsors/gearmax.webp",
    "tagline": "Level Up Your Gear",
    "tier": "GOLD",
    "displayDurationSeconds": 12,
    "displayOrder": 2
  }
]
```

ROTATION LOGIC (JavaScript):

└─ Array of sponsors with displayDurationSeconds

└─ Display each in order for its duration

└─ Loop continuously

- On OVERLAY_DATA_REFRESH: restart rotation with updated sponsor list
- On OVERLAY_COMMAND { command: "SHOW_SPONSOR", sponsorId: "uuid" }:
Skip to specific sponsor (for planned sponsor moments)

SECTION 11 — OVERLAY TYPE 06: WINNER OVERLAY

11.1 Overview

OVERLAY: OVL-06 – Winner Celebration (Full Screen)
Route: /overlay/tournament/{id}/winner
Size: 1920 × 1080 px (full screen)
Purpose: Epic winner reveal when tournament is COMPLETED
Updates: On TOURNAMENT_COMPLETED STOMP event
Trigger: Automatic on tournament completion OR manual producer trigger

11.2 Visual Layout Specification

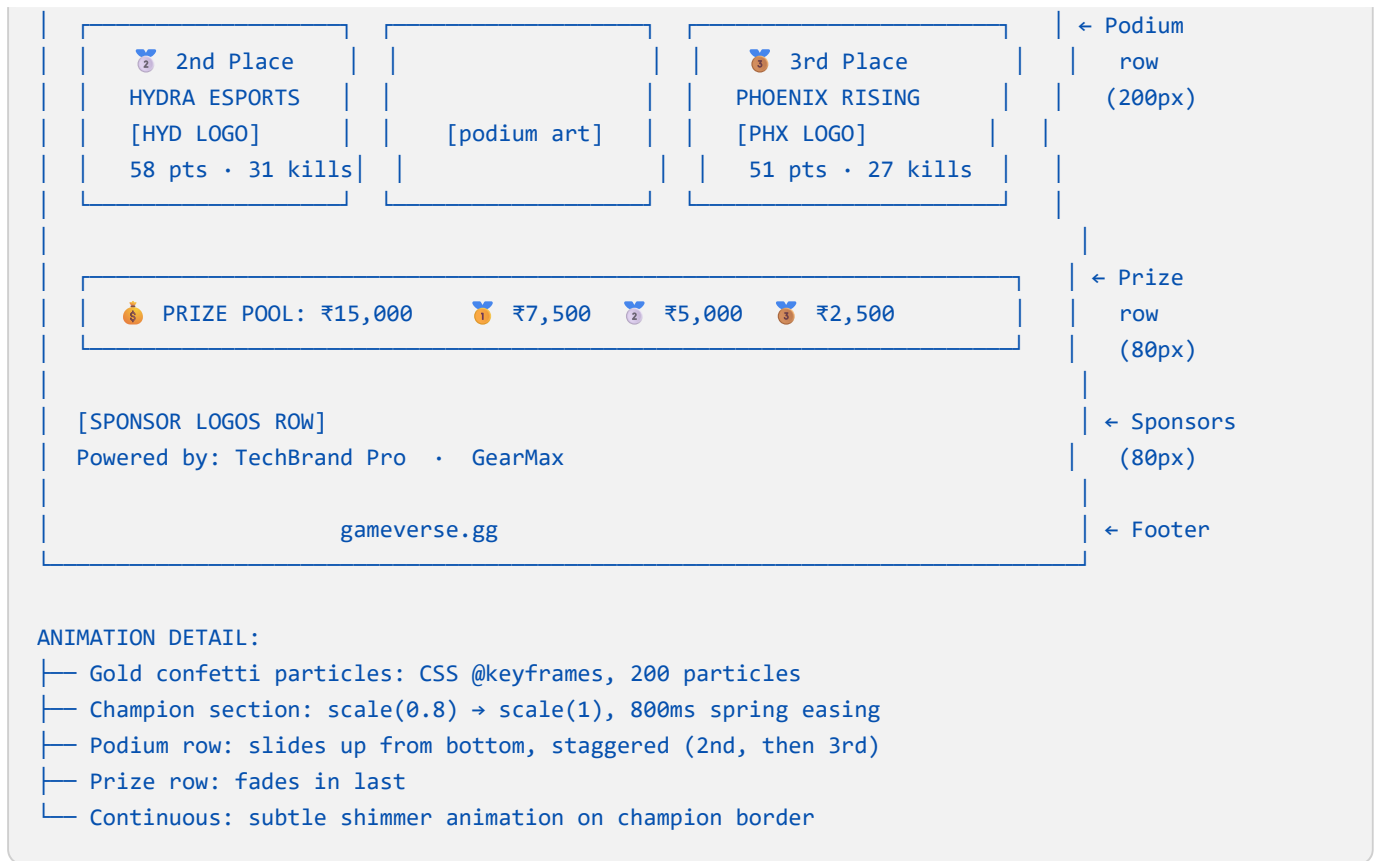
OVL-06: WINNER OVERLAY (1920 × 1080 px)

PHASE 1: ENTRY ANIMATION (3 seconds)

Gold particles rain from top
Tournament logo fades in center (scales 0 → 1, 800ms ease-out)
"TOURNAMENT COMPLETE" text slides up from below (500ms)

PHASE 2: CHAMPION REVEAL (shown until scene change)





11.3 Winner Data Schema

DATA CONSUMED FROM: TOURNAMENT_COMPLETED STOMP event + REST API fetch

window.__INITIAL_STATE__ when winner overlay loads:

```
{
  "tournament": {
    "id": "tid-001",
    "name": "BGMI Weekend Cup #12",
    "game": "BGMI",
    "prizePool": 15000.00,
    "currency": "INR"
  },
  "winners": [
    {
      "position": 1,
      "team": {
        "teamId": "team-uuid-stm",
        "teamName": "Storm Squad",
        "teamTag": "STM",
        "logoUrl": "https://cdn.gameverse.gg/teams/stm.webp",
        "logoUrlLarge": "https://cdn.gameverse.gg/teams/stm-lg.webp"
      },
      "stats": {
        "totalPoints": 62.0,
        "totalKills": 38,
        "chickenDinners": 2,
        "matchesPlayed": 16
      },
      "roster": [
```

```

    { "username": "ArjunOP", "displayName": "Arjun Sharma" },
    { "username": "RaviSnipe", "displayName": "Ravi Kumar" },
    { "username": "KiranRush", "displayName": "Kiran Singh" },
    { "username": "DevIGL", "displayName": "Dev Patel" }
  ],
  "prizeAmount": 7500.00
},
{
  "position": 2,
  "team": { ... },
  "prizeAmount": 5000.00
},
{
  "position": 3,
  "team": { ... },
  "prizeAmount": 2500.00
}
],
"sponsors": [ ... ],
"org": {
  "orgName": "Hydra Events",
  "logoUrl": "https://..."
}
}

```

11.4 Winner Overlay Variants

OVL-06B: WINNER PODIUM (Compact)

Route: /overlay/tournament/{id}/winner/podium

Size: 800 × 400 px

Purpose: Smaller winner reveal for corner or secondary placement

Content: Top 3 podium without full-screen celebration

Use: Side panel during final match reveal

OVL-06C: RESULT SPLASH (Per-Match)

Route: /overlay/tournament/{id}/result-splash

Size: 1920 × 400 px (horizontal banner)

Purpose: Shows top 3 results after EACH match (not just final)

Trigger: Automatic on RESULT_PUBLISHED (every match)

Auto-hide: After 15 seconds (configurable in Broadcast Dashboard)

OVL-06C VISUAL:

MATCH 3 RESULTS			Round 2		
1	STORM SQUAD	STM	Placement: 1st	Kills: 8	Points: 23
2	HYDRA ESPORTS	HYD	Placement: 2nd	Kills: 6	Points: 18
3	PHOENIX RISING	PHX	Placement: 3rd	Kills: 5	Points: 15

TRIGGER BEHAVIOR:

- RESULT_PUBLISHED event received
- Result splash slides in from top (400ms)
- Displays for 15 seconds

- Slides out to top (400ms)
- Automatically hides (CSS: display none)
- No producer action needed

Can be overridden:

- OVERLAY_COMMAND { command: HIDE, overlayType: RESULT_SPLASH }
Dismisses early if producer wants to switch scene
- OVERLAY_COMMAND { command: SHOW, overlayType: RESULT_SPLASH }
Re-shows if producer missed it

SECTION 12 — OVERLAY LIVE DATA UPDATE SYSTEM

12.1 Overlay Engine Architecture

OVERLAY ENGINE (overlay-engine.js):

Shared across ALL overlay types. Responsibilities:

- └ STOMP connection management
- └ Reconnection with exponential backoff
- └ Event routing to overlay-specific renderer
- └ Sequence number tracking (detect missed events)
- └ Connection status dot management
- └ Token refresh coordination

INITIALIZATION SEQUENCE:

1. Page loads → window.__INITIAL_STATE__ available
2. Overlay-type renderer.init(__INITIAL_STATE__) called
→ Renders initial state to DOM (instant, no flash)
3. overlay-engine.connect() called:
→ Create SockJS instance
→ Create STOMP Client
→ Connect with overlayToken in headers
4. On CONNECTED:
→ Subscribe to relevant STOMP topics
→ Status dot: ● green
5. Events received → dispatched to renderer
6. Renderer updates DOM with animations

STOMP SUBSCRIPTION PER OVERLAY TYPE:

Overlay Type	STOMP Subscriptions
Leaderboard (all)	/topic/tournament.{id}.leaderboard
	/topic/tournament.{id}.overlays
Match Info Bar	/topic/tournament.{id}.matches
	/topic/tournament.{id}.overlays
Team Statistics	/topic/tournament.{id}.leaderboard

	/topic/tournament.{id}.overlays
Player Statistics	/topic/tournament.{id}.leaderboard
	/topic/tournament.{id}.overlays
Sponsor Banner	/topic/tournament.{id}.overlays
	(static rotation, minimal WS needed)
Winner Overlay	/topic/tournament.{id}.status
	/topic/tournament.{id}.overlays
Result Splash	/topic/tournament.{id}.matches
	/topic/tournament.{id}.overlays

12.2 Event Handler Mapping

EVENT → OVERLAY ACTION MAPPING:

LEADERBOARD_UPDATED:

- └─ Leaderboard overlays (01, 01B, 01C):
 - updateRankedRows(event.data.leaderboard)
 - animateRankChanges(event.data.leaderboard)
- └─ Team Statistics overlay (03):
 - findFeaturedTeam(event.data.leaderboard, featuredTeamId)
 - updateTeamStats(featuredTeamData)
- └─ Player Statistics overlay (04):
 - findTeamOfPlayer(event.data.leaderboard, featuredPlayerId)
 - updatePlayerStats(playerData)

MATCH_STATUS_CHANGED:

- └─ Match Info Bar (02, 02B):
 - updateStatusBadge(event.data.currentStatus)
 - updateTimer(event.data.actualStartTime)
 - updateMatchInfo(event.data.matchNumber, roundNumber)
- └─ Result Splash (06C):
 - If status = COMPLETED: auto-hide splash
 - If status = IN_PROGRESS: clear previous result

RESULT_PUBLISHED:

- └─ Result Splash (06C):
 - buildResultSplash(event.data.results)
 - showSplash() → autoHideAfter(15000ms)
- └─ Team/Player Stats:
 - Update last match performance section

TOURNAMENT_COMPLETED:

- └─ Winner Overlay (06, 06B):
 - fetchWinnerData() via REST API
 - playChampionRevealAnimation()
 - renderWinnerPodium(winnerData)

OVERLAY_COMMAND:

- └─ All overlays:

```
switch(event.data.command) {
  case 'SHOW':          showOverlay()
  case 'HIDE':          hideOverlay()
  case 'UPDATE':        updateData(event.data.params)
  case 'REFRESH':       fetchFullData() → re-render
  case 'UPDATE_FEATURED': setFeaturedEntity(params.entityId)
  case 'SHOW_SPONSOR':  jumpToSponsor(params.sponsorId)
}
```

12.3 Live Update Latency by Overlay Type

LATENCY FROM EVENT TO VISIBLE OVERLAY CHANGE:

Overlay Type	Event Source	Target Latency
Leaderboard	LEADERBOARD_UPDATED (after result submit)	< 500ms (incl. animation)
Match Info Bar	MATCH_STATUS_CHANGED (status change)	< 300ms
Result Splash	RESULT_PUBLISHED (after verification)	< 400ms (slide animation)
Team Statistics	LEADERBOARD_UPDATED	< 500ms
Winner Overlay	TOURNAMENT_COMPLETED + REST API fetch	< 2000ms (data + animation)
Sponsor Banner	Static rotation	N/A (timer-based)

TOTAL FROM RESULT SUBMIT → OVERLAY VISIBLE:
Referee submits → DB → Leaderboard calc → STOMP → OBS Browser → DOM → Screen
~200ms (network + processing) + ~600ms (CSS animation) = ~800ms total
Well within 3-second target 

SECTION 13 — OVERLAY TOKEN & SECURITY

13.1 Token Lifecycle

OVERLAY TOKEN MANAGEMENT:

- GENERATION:
- Trigger: Broadcast Producer opens Overlay Management in Broadcast Dashboard
 - Spring Boot: Generate UUID v4 overlay token
 - Store in overlay_configs:

- | tournament_id
- | overlay_type
- | token_hash (SHA-256 – raw token never stored)
- | created_at
- | expires_at (tournament end date + 30 days)
- | is_active (true)
- └─ Raw token returned ONCE to Broadcast Dashboard (like a secret key)

TOKEN IN URL:

/overlay/tournament/{id}/leaderboard?token=550e8400-e29b-41d4-a716...

- └─ Token is a query parameter (not a path segment)
- └─ URL is used in OBS Browser Source settings

VALIDATION ON EVERY REQUEST:

GET /overlay/tournament/{id}/leaderboard?token=...

- └─ SHA-256 hash the token from query param
- └─ Lookup in overlay_configs WHERE token_hash = ? AND tournament_id = ?
- └─ Check: is_active = true
- └─ Check: expires_at > NOW()
- └─ Check: overlay_type matches URL path
- └─ If valid: Serve overlay HTML with initial state injected
- └─ If invalid: Return 401 HTML error page (not blank, shows error message)

TOKEN ROTATION:

Broadcast Producer → Broadcast Dashboard → Rotate Token button

- └─ Old token: is_active = false (immediate)
- └─ New token: generated
- └─ Producer updates OBS Browser Source URL with new token
- └─ Old OBS source: Next heartbeat check fails → disconnects
- └─ New OBS source: Connects and works normally

WHAT TOKEN GRANTS:

- └─ Serve overlay HTML page ✓
- └─ STOMP subscribe to public tournament topics ✓
- └─ STOMP subscribe to /topic/tournament.{id}.overlays ✓
- └─ No access to: command topics, user queues, admin APIs ✗

TOKEN SECURITY PROPERTIES:

- └─ Compromise risk: LOW (overlay data is mostly public leaderboard data)
- └─ Mitigation: Instant rotation if compromised
- └─ No PII exposed: Overlay never shows player UUIDs, emails, phone numbers

SECTION 14 — OVERLAY CUSTOMIZATION & THEMING

14.1 Theme System

THEMING ARCHITECTURE:

Overlays support two theming approaches:

APPROACH 1: Organization Brand Kit (Automatic)

Org Owner/Admin configures brand in Organization Settings:

- └ Primary color: #6B48FF (purple)
- └ Secondary color: #FF4B6E (coral)
- └ Accent color: #FFD700 (gold)
- └ Background: #0A0A1A (near black)
- └ Text color: #FFFFFF
- └ Font family: "Rajdhani" (or custom)
- └ Logo URL: org logo for overlay header

These inject as CSS variables into every overlay:

```
:root {  
  --brand-primary: #6B48FF;  
  --brand-secondary: #FF4B6E;  
  --brand-accent: #FFD700;  
  --brand-bg: #0A0A1A;  
  --brand-text: #FFFFFF;  
}
```

All overlay CSS uses var(--brand-primary) etc.

Change org colors → all overlays update on next load.

APPROACH 2: URL Theme Parameter

- ?theme=dark → Default dark esports theme
- ?theme=light → Light background theme
- ?theme=minimal → Clean minimal (less glassmorphism)
- ?theme=custom → Uses ONLY org brand variables (no defaults)

APPROACH 3: Scale Parameter

- ?scale=1.0 → Default (1080p)
 - ?scale=1.25 → Slightly larger text
 - ?scale=1.5 → Large (for 1440p streams)
 - ?scale=2.0 → Double size (for 4K streams)
- All px values multiplied by scale factor via CSS transform

14.2 Visual Design Specifications

DESIGN LANGUAGE – GAMEVERSE OVERLAYS:

AESTHETIC: Dark esports with glassmorphism accents

BASE COLORS (default dark theme):

- └ Background: rgba(10, 10, 26, 0.90) (near-black, 90% opacity)
- └ Glass panels: rgba(255, 255, 255, 0.05) (frosted glass effect)
- └ Borders: rgba(255, 255, 255, 0.10)
- └ Text primary: #FFFFFF
- └ Text secondary: #A0AEC0 (muted grey)
- └ Text accent: var(--brand-primary)

EFFECTS:

- └ Backdrop blur: backdrop-filter: blur(10px) on panels

- └─ Box shadows: 0 4px 16px rgba(0,0,0,0.4) on cards
- └─ Borders: 1px solid rgba(255,255,255,0.08)
- └─ Gradients: Linear gradient from brand-primary to brand-secondary on rank #1 row, header, etc.

RANK COLOR CODING:

- └─ Rank 1: Gold gradient background → #FFD700 accent
- └─ Rank 2: Silver tint → #C0C0C0 accent
- └─ Rank 3: Bronze tint → #CD7F32 accent
- └─ Ranks 4+: Default glass panel
- └─ DQ teams: Red tint + strikethrough

RANK CHANGE COLORS:

- └─ Moved up (▲): #10B981 (emerald green)
- └─ Moved down (▼): #EF4444 (red)
- └─ No change (-): #6B7280 (grey)

TYPOGRAPHY:

- └─ Tournament name: Rajdhani Bold, 24px
- └─ Team names: Rajdhani SemiBold, 18px
- └─ Team tags: Rajdhani Regular, 14px, muted
- └─ Points/stats: JetBrains Mono, 16px (monospace for alignment)
- └─ Headers: Rajdhani Bold, 12px, uppercase, letter-spacing: 2px
- └─ Footer/timestamps: Inter, 11px, muted

SECTION 15 — BROADCAST PRODUCER DASHBOARD

15.1 Dashboard Overview

BROADCAST PRODUCER DASHBOARD

(React component within Command Center: /command-center/{id}/broadcast)

SECTIONS:

BROADCAST DASHBOARD — BGMI Weekend Cup #12	
SECTION A: STREAM STATUS	SECTION B: OVERLAY MANAGEMENT
Platform: YouTube	OVL-01 Leaderboard Connected
Bitrate: 6000kbps	OVL-02 Match Bar Connected
FPS: 60	OVL-05 Sponsor Connected
Dropped: 0.02%	OVL-06 Winner Standby
CPU: 47%	[Configure] [Copy URL] [Rotate Token]
Viewers: 1,204	
Peak: 1,842	
SECTION C: OBS CONTROL	SECTION D: LEADERBOARD PREVIEW
OBS: Connected	[Live preview of leaderboard overlay]
Current Scene: Match Live	1. Storm Squad STM 62pts

Quick Scenes: [Match Live] [Break Screen] [Result Show] [Grand Finale]	2. Hydra Esports HYD 58pts 3. Phoenix Rising PHX 51pts ...
SECTION E: OVERLAY COMMANDS	SECTION F: STREAM ANNOTATIONS
Result Splash: [Show][Hide] Team Spotlight: [Select ▼] Player Spotlight: [Select ▼] Sponsor: [Jump to ▼] Winner Screen: [Trigger]	[⚡ Match Start] [🎮 Match End] [🍗 Chicken Dinner] [💬 Custom] Recent: 14:30:05 - Match 3 Start 14:42:18 - Chicken Dinner (Storm) 14:44:55 - Match 3 End

15.2 Overlay Management Panel

OVERLAY MANAGEMENT (SECTION B – DETAILED):

For each overlay type:

LEADERBOARD OVERLAY (OVL-01)

CONNECTED

Browser Source URL:
<https://gameverse.gg/overlay/tournament/tid-001/leaderboard?token=550e8400-e29b-41d4-a716-446655440000>

Copy URL
 Rotate Token

OBS Settings: Width: 420px Height: 900px

Copy Settings

Configuration:

Theme: [Dark ▼]

Scale: [1.0 ▼]

Show top: [16 teams ▼]

Show kills column: [☒ Yes]

Show CD column: [☒ Yes]

Animate rank changes:[☒ Yes]

Save Configuration

Connected browsers: 1 (OBS Browser Source)
 Last update received: 2 seconds ago
 Preview: [Small leaderboard thumbnail showing live data]

Actions: [Preview Full] [Force Refresh] [Advanced]

15.3 OBS Scene Control

OBS SCENE CONTROL (SECTION C – DETAILED):

Requires OBS WebSocket connection to producer's OBS instance.

Connection Setup:

- └─ Producer opens OBS → Tools → WebSocket Server Settings → Enable
- └─ Port: 4455, Password: [set by producer]
- └─ In Broadcast Dashboard: Enter ws://localhost:4455 + password
- └─ Click Connect → "● OBS Connected – 8 scenes detected"

Quick Scene Buttons:

- [Match Live] → OBS scene: "Match Live" (game capture + leaderboard + bar)
- [Break Screen] → OBS scene: "Break Screen" (graphic + sponsor + countdown)
- [Result Show] → OBS scene: "Result Show" (game + full leaderboard)
- [Grand Finale] → OBS scene: "Grand Finale" (winner overlay full screen)
- [Player Spot] → OBS scene: "Player Spotlight" (game + player card)

Auto-Trigger (configurable):

- └─ RESULT_PUBLISHED → Auto switch to "Result Show" scene
 - └─ Producer setting: [☒ Auto-switch on result]
 - └─ Duration: Switch back to "Match Live" after [30 ▼] seconds
- └─ TOURNAMENT_COMPLETED → Auto switch to "Grand Finale" scene
 - └─ Producer setting: [☒ Auto-switch on tournament complete]

Implementation:

- Spring Boot → OVERLAY_COMMAND { command: SCENE_CHANGE, scene: "Match Live" }
- Broadcast Dashboard React component receives STOMP event
- Calls OBS WebSocket API: SetCurrentProgramScene
- OBS switches scene
- Dashboard shows: "☒ Scene switched to Match Live"

SECTION 16 — OBS WEBSOCKET INTEGRATION

16.1 OBS WebSocket Protocol

OBS WEBSOCKET INTEGRATION:

OBS Studio 28+ includes OBS WebSocket server (obs-websocket 5.x protocol).
GameVerse Broadcast Dashboard connects to producer's local OBS instance.

CONNECTION:

Producer's OBS Machine:

- └─ OBS Studio running on Windows/Mac/Linux
 - └─ WebSocket server: ws://localhost:4455
 - └─ Password: set by producer in OBS settings

GameVerse Broadcast Dashboard (React):

- └─ Connects to ws://localhost:4455 via WebSockets API

(Note: Browser connects to localhost – requires producer on same machine
OR proxy configuration for remote OBS)

OPERATIONS USED:

1. GetSceneList
 - List all configured OBS scenes
 - Displayed as quick-switch buttons in dashboard
 - Used to validate scene names
2. SetCurrentProgramScene
 - Switch OBS to named scene
 - Triggered by: Producer button OR auto-trigger on STOMP event
3. GetStreamStatus
 - Check if stream is live, bitrate, FPS, dropped frames
 - Polled every 5 seconds for stream health display
4. StartStream / StopStream
 - Start/stop stream from dashboard (convenience)
 - Confirmation dialog before stopping
5. GetStats
 - CPU usage, memory usage, rendering lag
 - Displayed in stream health section

NOTE ON ARCHITECTURE:

OBS WebSocket is browser-to-OBS, not server-to-OBS.
Spring Boot does NOT directly control OBS.
The React Broadcast Dashboard (running in producer's browser)
connects directly to local OBS WebSocket.
Spring Boot sends STOMP events → React receives →
React calls OBS WebSocket API.

SECTION 17 — STREAM HEALTH MONITORING

17.1 Stream Health Dashboard

STREAM HEALTH INDICATORS:

DATA SOURCES:

- └ OBS WebSocket: GetStreamStatus + GetStats (every 5 seconds)
- └ YouTube/Twitch API: Viewer count (every 60 seconds)
- └ STOMP: STREAM_HEALTH_UPDATE event (pushed by server when fetched)

HEALTH METRICS DISPLAYED:

Metric	Good	Warning	Critical
Bitrate	> 4500kbps	2000–4500kbps	< 2000kbps

FPS	> 55fps	45-55fps	< 45fps
Dropped Frames	< 0.5%	0.5-2%	> 2%
CPU Usage	< 70%	70-85%	> 85%
OBS Connection	Connected	Reconnecting	Disconnected
Stream Status	Live	Starting	Offline

VISUAL INDICATORS:

- └─ ● Green dot = Good (all metrics in "Good" range)
- └─ ● Yellow dot = Warning (any metric in "Warning" range)
- └─ ● Red dot = Critical (any metric in "Critical" range)
- └─ ● Grey dot = Stream offline / OBS not connected

ALERTS:

- └─ Bitrate drops below 2000kbps: Toast alert (P1)
- └─ Dropped frames exceeds 2%: Toast alert (P1)
- └─ OBS disconnects: Prominent banner + notification (P0)
- └─ Stream goes offline unexpectedly: Immediate alert (P0)

SECTION 18 — OVERLAY ANIMATION SPECIFICATION

18.1 Complete Animation Reference

ALL ANIMATIONS USED ACROSS OVERLAYS:

LEADERBOARD ANIMATIONS:

rankRowSlide:

- Property: transform: translateY
- Duration: 600ms
- Easing: cubic-bezier(0.4, 0, 0.2, 1) (Material ease)
- Trigger: Rank order changes after LEADERBOARD_UPDATED

rankUp:

- Property: background-color flash (transparent → green → transparent)
- Duration: 1200ms
- Keyframes: 0% → transparent | 30% → rgba(16,185,129,0.3) | 100% → transparent
- Trigger: Team rank improves

rankDown:

- Property: background-color flash (transparent → red → transparent)
- Duration: 1200ms
- Keyframes: 0% → transparent | 30% → rgba(239,68,68,0.2) | 100% → transparent
- Trigger: Team rank drops

pointsUpdate:

- Property: color flash on points value (white → gold → white)
- Duration: 800ms
- Trigger: Points value changes for a team

chickenDinner:

Property: gold border glow + star spin
Duration: 1500ms (border) + 500ms (star spin)
Trigger: Team achieves chicken dinner (1st place)

MATCH INFO BAR ANIMATIONS:

statusPulse (IN_PROGRESS):

Property: opacity 1 → 0.4 → 1
Duration: 1500ms, infinite
Trigger: Match status = IN_PROGRESS

statusPulseOrange (PAUSED):

Property: opacity + background-color pulse orange
Duration: 1000ms, infinite
Trigger: Match status = PAUSED

timerTick:

Property: No animation (just text update every second)
Format: MM:SS counting up from match start

RESULT SPLASH ANIMATIONS:

splashSlideIn:

Property: transform: translateY(-100%) → translateY(0)
Duration: 400ms
Easing: cubic-bezier(0.34, 1.56, 0.64, 1) (spring overshoot)
Trigger: RESULT_PUBLISHED event

splashSlideOut:

Property: transform: translateY(0) → translateY(-100%)
Duration: 300ms
Easing: ease-in
Trigger: After displayDurationSeconds OR OVERLAY_COMMAND HIDE

WINNER OVERLAY ANIMATIONS:

confettiRain:

Type: CSS @keyframes on 200 particle divs
Duration: 3000ms initial burst + continuous subtle fall
Colors: Brand primary, gold, white, silver

championReveal:

Property: transform: scale(0.8) opacity(0) → scale(1) opacity(1)
Duration: 800ms
Easing: cubic-bezier(0.34, 1.56, 0.64, 1) (spring)
Delay: 1500ms (after confetti starts)

podiumSlideUp:

Property: transform: translateY(100%) → translateY(0)
Duration: 500ms
Easing: ease-out
Delay: 2nd place: 2500ms | 3rd place: 2800ms

shimmerBorder:

Type: CSS @keyframes on champion card border
Property: background-position (gradient slide)

Duration: 2000ms, infinite
Effect: Gold shimmer animating along border

SPONSOR BANNER ANIMATIONS:

sponsorFadeOut:

Property: opacity: 1 → 0
Duration: 500ms
Trigger: Timer expires for current sponsor

sponsorFadeIn:

Property: opacity: 0 → 1
Duration: 500ms
Trigger: After previous sponsor fades out

PERFORMANCE NOTES:

- └ All animations use transform and opacity only (GPU-accelerated)
- └ No layout-triggering properties animated (no width, height, top, left)
- └ will-change: transform applied to animated elements
- └ 60fps maintained on OBS Chromium (tested requirement)

SECTION 19 — TESTING & QA FOR OVERLAYS

19.1 Testing Strategy

OVERLAY TESTING APPROACH:

UNIT TESTS (Vitest + JSDOM):

- └ Leaderboard renderer:
 - └ Renders correct number of rows (16, 10, 5)
 - └ Correct rank order (highest points first)
 - └ Rank change direction calculated correctly
 - └ DQ teams shown at bottom with DQ badge
 - └ Points formatted correctly (no decimals for whole numbers)
- └ Match info bar:
 - └ Status badge shows correct color per status enum
 - └ Timer counts up correctly from actualStartTime
 - └ Countdown shows correct time to scheduled start
- └ Result splash:
 - └ Shows top 3 from results array (sorted by placement)
 - └ Auto-hides after displayDurationSeconds
 - └ OVERLAY_COMMAND HIDE dismisses early

INTEGRATION TESTS (Spring Boot Test):

- └ Overlay route: GET /overlay/tournament/{id}/leaderboard
 - └ Valid token → 200 with HTML containing __INITIAL_STATE__
 - └ Invalid token → 401 HTML error page
 - └ Expired token → 401 HTML error page

- └─ __INITIAL_STATE__ contains correct leaderboard data
- └─ Overlay STOMP connection:
 - └─ Connect with valid overlay token → CONNECTED
 - └─ Subscribe to /topic/tournament.{id}.leaderboard → allowed
 - └─ Subscribe to /topic/tournament.{id}.command → denied
 - └─ Cannot subscribe to /user/*/queue/* with overlay token

VISUAL REGRESSION TESTS (Playwright):

- └─ Screenshot each overlay with test tournament data
- └─ Compare to baseline screenshots (pixel diff < 0.1%)
- └─ Test overlay on different viewport sizes (scale parameter)
- └─ Test both dark and light themes
- └─ Test each animation with video recording

E2E BROADCAST TESTS (Playwright with OBS simulation):

- └─ Submit match result
 - Assert: Leaderboard overlay updates within 3 seconds
 - Assert: Rank change animations play (CSS class check)
 - Assert: Result splash appears automatically
 - Assert: Result splash disappears after 15 seconds
- └─ Tournament complete
 - Assert: Winner overlay displays champion team name
 - Assert: Confetti animation plays (class check)
 - Assert: All top 3 teams shown correctly
- └─ Reconnection
 - Kill STOMP connection
 - Assert: Connection dot turns red (● → ●)
 - Wait for reconnect (5 seconds)
 - Assert: Connection dot turns green
 - Assert: Leaderboard shows current data (not stale)
- └─ Overlay token rotation
 - Rotate token in Broadcast Dashboard
 - Assert: Old overlay disconnects (dot goes grey)
 - Load overlay with new token URL
 - Assert: New overlay connects and shows correct data

SECTION 20 — TROUBLESHOOTING GUIDE

20.1 Common Issues & Resolutions

TROUBLESHOOTING GUIDE FOR BROADCAST PRODUCERS:

ISSUE: Overlay shows blank/white background in OBS

Cause: OBS Browser Source custom CSS overriding transparent bg

Fix: Clear the "Custom CSS" field in OBS Browser Source settings
The overlay handles its own background transparency.

ISSUE: Overlay shows "401 - Invalid Token" error

Cause A: Token was rotated – URL is outdated

Fix: Go to Broadcast Dashboard → Overlays → Copy new URL
Update OBS Browser Source with new URL

Cause B: Tournament ended > 30 days ago – token expired

Fix: Contact GameVerse support if needed for post-event access

ISSUE: Connection status dot is  Red (disconnected)

Cause A: OBS production machine lost internet

Fix: Check internet connection. Overlay auto-reconnects when restored.
Data is frozen at last known state – safe to continue broadcasting.

Cause B: GameVerse server restart/maintenance

Fix: Wait 30-60 seconds. Overlay reconnects automatically.
Status page: <https://status.gameverse.gg>

ISSUE: Leaderboard not updating after result submitted

Cause A: Result not yet published (pending director verification)

Check: Command Center → Scoring → Result status = PENDING_VERIFICATION

Fix: Director must verify and approve the result

Cause B: Overlay disconnected from WebSocket

Check: Connection status dot is red/grey

Fix: Refresh OBS Browser Source (right-click → Refresh)
Or click "Force Refresh" in Broadcast Dashboard

ISSUE: OBS WebSocket shows "Not Connected" in Broadcast Dashboard

Check 1: OBS Studio is running on the production machine

Check 2: OBS → Tools → WebSocket Server Settings → Server Enabled 

Check 3: Port 4455 is not blocked by firewall

Check 4: Password in Broadcast Dashboard matches OBS WebSocket password

Fix: Verify all above. Firewall: allow port 4455 on loopback (localhost).

Note: OBS WebSocket only works when dashboard open on SAME machine as OBS.

ISSUE: Overlay text too small / too large for stream

Fix: Add scale parameter to overlay URL:

?scale=1.5 → 50% larger

?scale=2.0 → Double size (for 4K)

OR: Adjust OBS Browser Source dimensions proportionally

ISSUE: Winner overlay did not appear on tournament completion

Cause A: Winner overlay not added as Browser Source in OBS

Fix: Add /overlay/tournament/{id}/winner as Browser Source
 Switch to Grand Finale scene that contains this source

Cause B: Auto-trigger not configured

Fix: Broadcast Dashboard → OBS Control →  Auto-switch on complete

Cause C: Tournament completed but winner overlay in wrong OBS scene

Fix: Manually click [Grand Finale] scene button in Broadcast Dashboard
 OR drag winner overlay Browser Source into correct scene in OBS

ISSUE: Sponsor rotation stopped / showing wrong sponsor

Fix: Broadcast Dashboard → Overlay Commands → Sponsor → [Force Refresh]
 This sends OVERLAY_DATA_REFRESH command → sponsor overlay reloads list

20.2 Quick Diagnostic Checklist

PRE-TOURNAMENT BROADCAST CHECKLIST:

T-60 minutes before first match:

OVERLAYS:

- ☐ All Browser Sources added to OBS scenes
- ☐ All overlays showing  Connected in Broadcast Dashboard
- ☐ Leaderboard preview shows registered teams (alphabetical pre-tournament)
- ☐ Match Info Bar shows "SCHEDULED" status for Match 1
- ☐ Sponsor banner showing correct sponsors and rotating
- ☐ Winner overlay added to Grand Finale scene (hidden/standby)

OBS:

- ☐ OBS WebSocket connected in Broadcast Dashboard
- ☐ All 4+ scenes configured (Match Live, Break, Result, Finale)
- ☐ Stream configured (key, platform URL)
- ☐ Test stream to confirm bitrate/quality

BROADCAST DASHBOARD:

- ☐ Stream health shows green for all metrics
- ☐ Viewer count polling working (shows 0 pre-stream)
- ☐ OBS scene quick-switch buttons tested
- ☐ Annotation tools tested (dummy annotation added + deleted)

TOURNAMENT:

- ☐ All team registrations APPROVED
- ☐ Match schedule published
- ☐ Check-in window timing confirmed with organizer
- ☐ First match start time confirmed

COMMUNICATION:

- ☐ Contact channel with Tournament Director confirmed (Discord/WhatsApp)

- ☐ Contact with Referees confirmed
- ☐ Escalation contact for tech issues confirmed

DOCUMENT COMPLETION SUMMARY

Overlay Coverage

Overlay	Route	Auto-Update	Animation	Status
OVL-01: Leaderboard Full	<code>/leaderboard</code>	✓ LEADERBOARD_UPDATED	✓ Row slide + rank flash	✓ Defined
OVL-01B: Leaderboard Top10	<code>/leaderboard/top10</code>	✓	✓	✓ Defined
OVL-01C: Leaderboard Top5	<code>/leaderboard/top5</code>	✓	✓	✓ Defined
OVL-02: Match Info Bar	<code>/match-info</code>	✓ MATCH_STATUS_CHANGED	✓ Status pulse	✓ Defined
OVL-02B: Match Countdown	<code>/match-countdown</code>	✓	✓ Red pulse < 60s	✓ Defined
OVL-03: Team Statistics	<code>/team/{id}/stats</code>	✓ RESULT_PUBLISHED	✓ Data flash	✓ Defined
OVL-03B: Team vs Team	<code>/team/compare</code>	✓	✓ Bar animation	✓ Defined
OVL-04: Player Statistics	<code>/player/{id}/stats</code>	✓ RESULT_PUBLISHED	✓ Data flash	✓ Defined
OVL-04B: Kill Feed	<code>/player/killfeed</code>	✓ Manual + STOMP	✓ Slide in	✓ Defined
OVL-05: Sponsor Banner	<code>/sponsor</code>	Static rotation	✓ Fade transition	✓ Defined
OVL-05B: Sponsor Corner	<code>/sponsor/corner</code>	Static	None	✓ Defined
OVL-06: Winner Full	<code>/winner</code>	✓ TOURNAMENT_COMPLETED	✓ Confetti + reveal	✓ Defined
OVL-06B: Winner Podium	<code>/winner/podium</code>	✓	✓ Podium slide	✓ Defined
OVL-06C: Result Splash	<code>/result-splash</code>	✓ RESULT_PUBLISHED	✓ Slide in/out	✓ Defined

Document Statistics

Element	Count
Overlay Types Defined	14
Overlay Routes Defined	14
Animation Types Specified	15+
STOMP Events Consumed	6
OBS Operations Used	5
Troubleshooting Scenarios	8
Pre-Tournament Checklist Items	20

Document Sign-Off

Role	Name	Status
Product Lead	—	Pending Review
Engineering Lead	—	Pending Review
Broadcast / Design Lead	—	Pending Review
QA Lead	—	Pending Review

Document: GameVerse OBS & Broadcast Integration Specification | **Version:** 1.0 | **Overlays:** 14 Types

| Status: Complete 